Spectral Simulation Report

Automated Analysis

June 18, 2025

Contents

1 Introduction

This report presents the results of spectral simulations derived from HDF5 dataset analysis. Each sample includes a spectral plot displayed in a full horizontal page and a detailed table summarizing the initial composition, actual positions, total pixels, and signal-to-noise ratio (SNR), along with simulation parameters such as temperature and electron density.

2 Sample: 11469

2.1 Simulation Parameters

Parameter	**Value**
Temperature (T)	8838 K
Electron Density (n_e)	2.78e+14 $\mathrm{cm}^{-}3$
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	33.36

Table 1: Simulation parameters and additional metrics for sample 11469.

2.2 Composition Analysis

Element	**Initial Composition (%)**	**Actual Positions**
Al	2.61	399.0
Ar	0.00	0.0
C	0.00	0.0
Ca	2.65	150.0
Cl	7.36	35.0
Co	5.62	1404.0
Cr	5.09	397.0
Fe	8.39	2738.0
Mg	11.92	343.0
Mn	10.31	706.0
N	11.43	401.0
Na	6.99	322.0
Ni	9.78	478.0
O	7.01	152.0
S	4.43	207.0
Si	3.73	220.0
Ti	2.66	1183.0
background	N/A	536.0
Total Positions		9671.0

Table 2: Composition analysis for sample 11469.

2.3 Spectral Plot

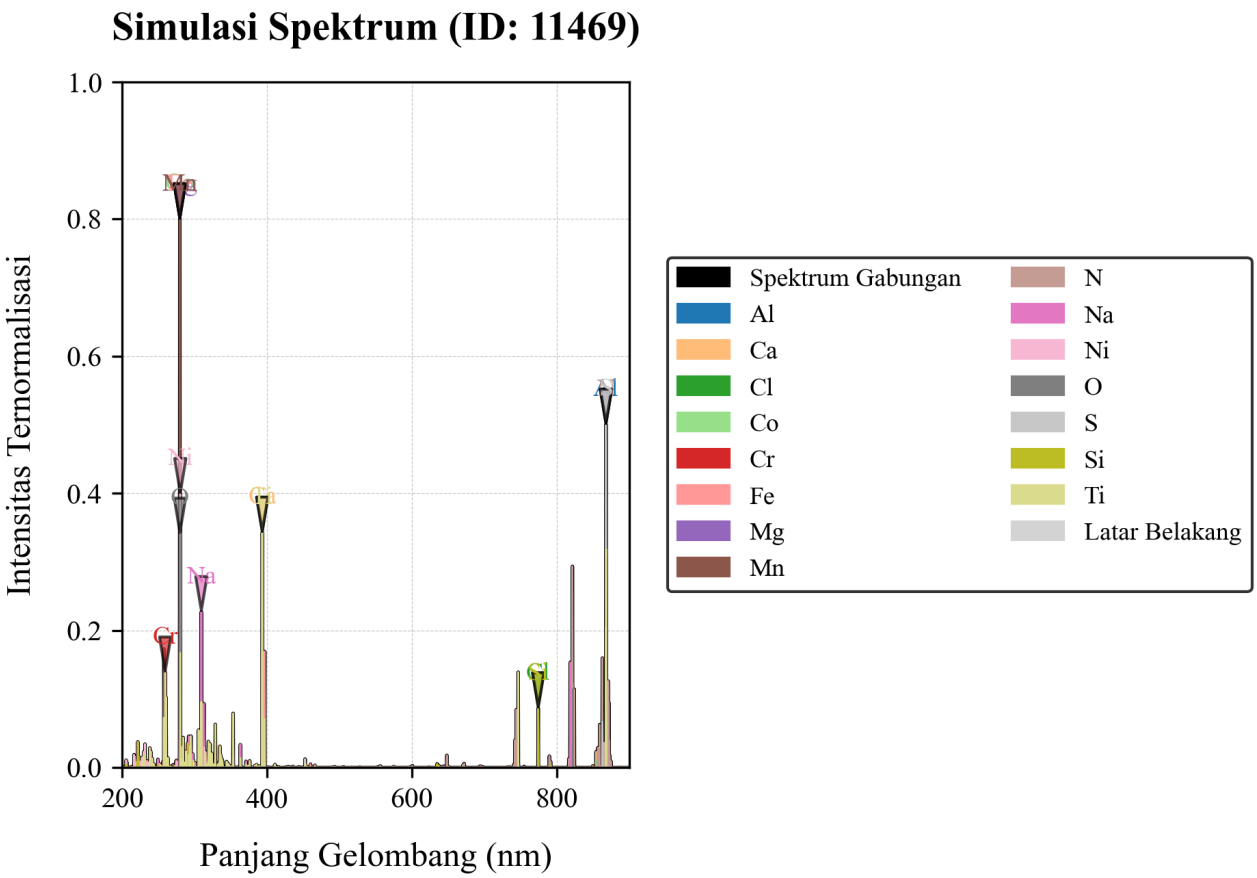

Figure 1: Spectral simulation plot for sample 11469 in full horizontal layout.

3 Sample: 9375

3.1 Simulation Parameters

Parameter	Value
Temperature (T)	10657 K
Electron Density (n_e)	1.38e+16 $\mathrm{cm}^{-}3$
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	37.77

Table 3: Simulation parameters and additional metrics for sample 9375.

3.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	0.00	0.0
Ar	0.00	0.0
C	11.20	330.0
Ca	14.63	150.0
Cl	1.84	31.0
Co	7.18	1404.0
Cr	2.10	397.0
Fe	11.57	2762.0
Mg	14.76	356.0
Mn	4.25	706.0
N	0.00	0.0
Na	3.38	322.0
Ni	14.01	478.0
O	0.00	0.0
S	4.69	228.0
Si	10.38	223.0
Ti	0.00	0.0
background	N/A	657.0
Total Positions		8044.0

Table 4: Composition analysis for sample 9375.

3.3 Spectral Plot

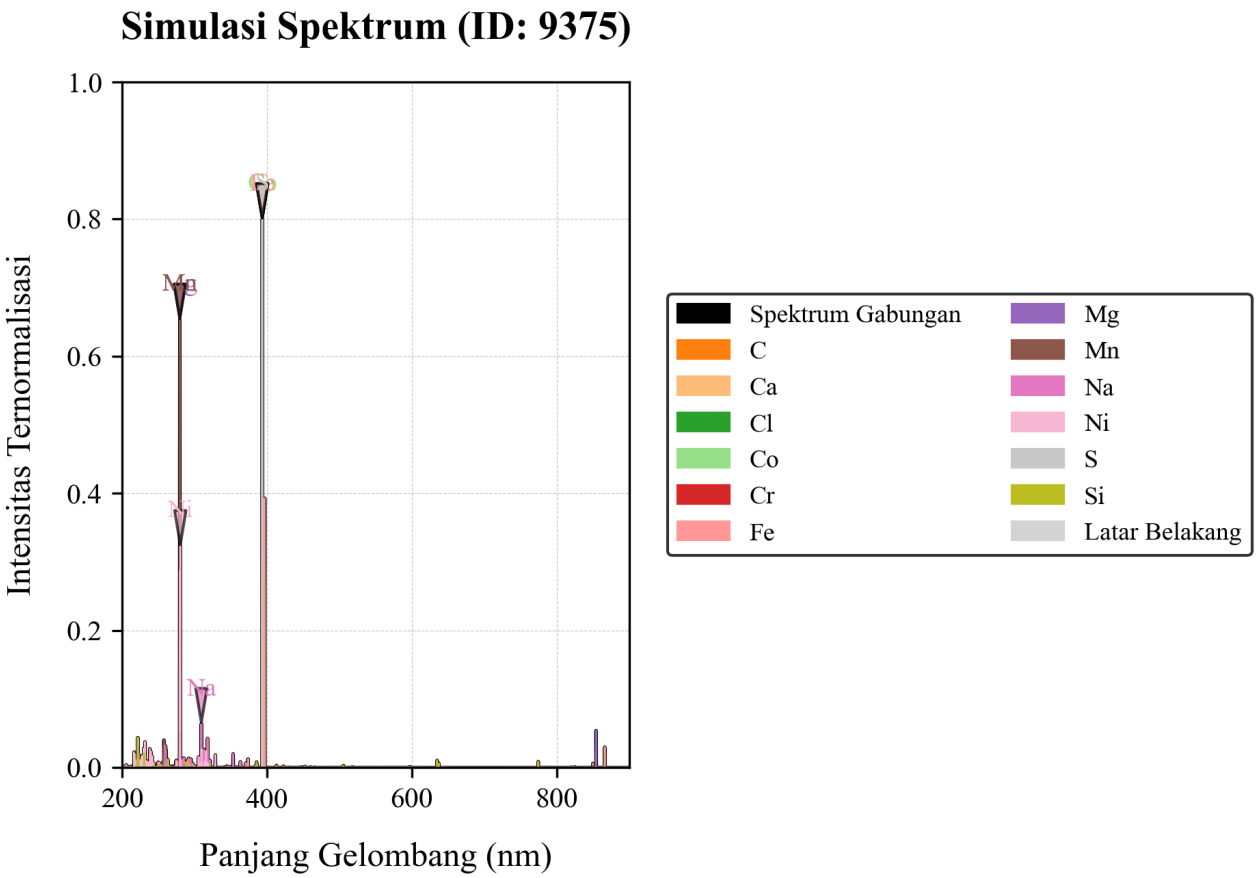

Figure 2: Spectral simulation plot for sample 9375 in full horizontal layout.

4 Sample: 5579

4.1 Simulation Parameters

Parameter	Value
Temperature (T)	6616 K
Electron Density (n_e)	1.56e+17 $\mathrm{cm}^{-}3$
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	39.94

Table 5: Simulation parameters and additional metrics for sample 5579.

4.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	4.93	158.0
Ar	6.15	286.0
C	7.92	209.0
Ca	10.45	150.0
Cl	10.18	12.0
Co	11.41	866.0
Cr	8.33	397.0
Fe	0.00	0.0
Mg	2.98	161.0
Mn	5.20	655.0
N	4.46	396.0
Na	11.70	330.0
Ni	2.17	478.0
O	7.67	50.0
S	1.32	25.0
Si	2.97	168.0
Ti	2.14	1181.0
background	N/A	1199.0
Total Positions		6721.0

Table 6: Composition analysis for sample 5579.

4.3 Spectral Plot

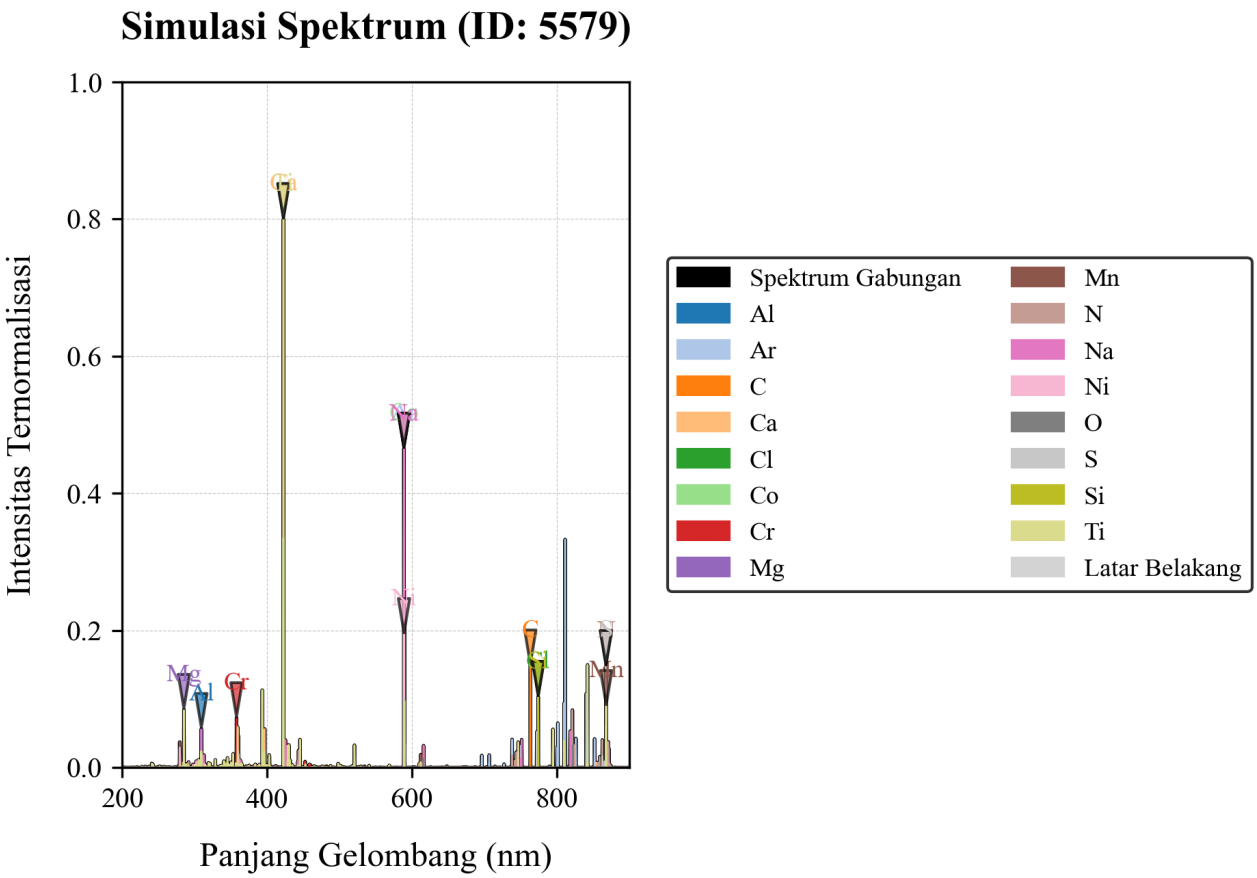

Figure 3: Spectral simulation plot for sample 5579 in full horizontal layout.

5 Sample: 17647

5.1 Simulation Parameters

Parameter	Value
Temperature (T)	6212 K
Electron Density (n_e)	1.48e+15 $\mathrm{cm}^{-}3$
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	37.77

Table 7: Simulation parameters and additional metrics for sample 17647.

5.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	0.00	0.0
Ar	12.05	408.0
C	0.00	0.0
Ca	10.73	150.0
Cl	1.21	12.0
Co	0.00	0.0
Cr	16.37	397.0
Fe	0.00	0.0
Mg	10.56	184.0
Mn	15.40	684.0
N	0.00	0.0
Na	4.13	322.0
Ni	6.04	478.0
O	0.00	0.0
S	11.97	25.0
Si	11.54	208.0
Ti	0.00	0.0
background	N/A	2102.0
Total Positions		4970.0

Table 8: Composition analysis for sample 17647.

5.3 Spectral Plot

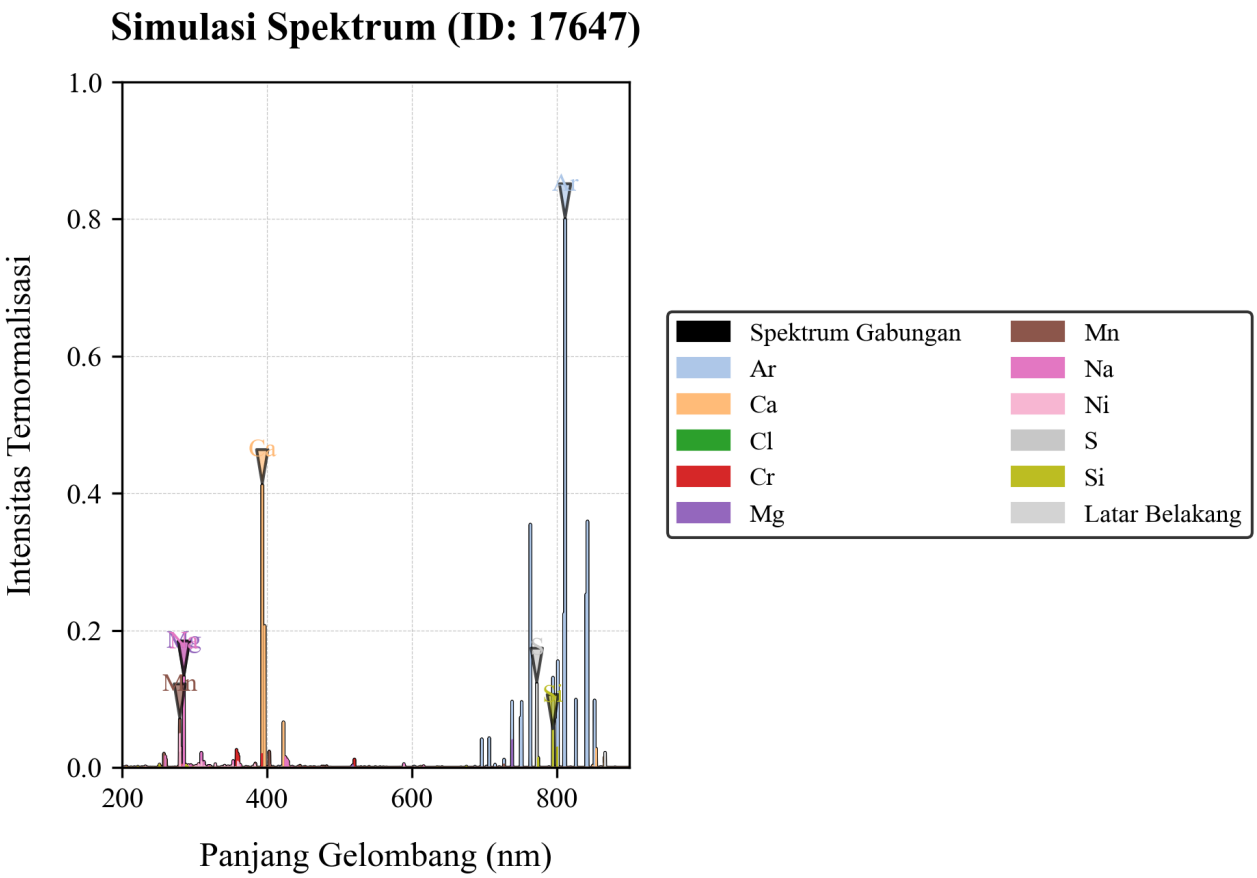

Figure 4: Spectral simulation plot for sample 17647 in full horizontal layout.

6 Sample: 17016

6.1 Simulation Parameters

Parameter	Value
Temperature (T)	5606 K
Electron Density (n_e)	4.23e+16 $\mathrm{cm}^{-}3$
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	36.92

Table 9: Simulation parameters and additional metrics for sample 17016.

6.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	2.95	128.0
Ar	9.73	246.0
C	0.00	0.0
Ca	11.22	150.0
Cl	9.87	12.0
Co	8.29	795.0
Cr	12.37	397.0
Fe	0.00	0.0
Mg	0.68	135.0
Mn	7.21	552.0
N	0.00	0.0
Na	7.37	330.0
Ni	9.41	478.0
O	5.47	8.0
S	5.15	25.0
Si	10.28	128.0
Ti	0.00	0.0
background	N/A	1995.0
Total Positions		5379.0

Table 10: Composition analysis for sample 17016.

6.3 Spectral Plot

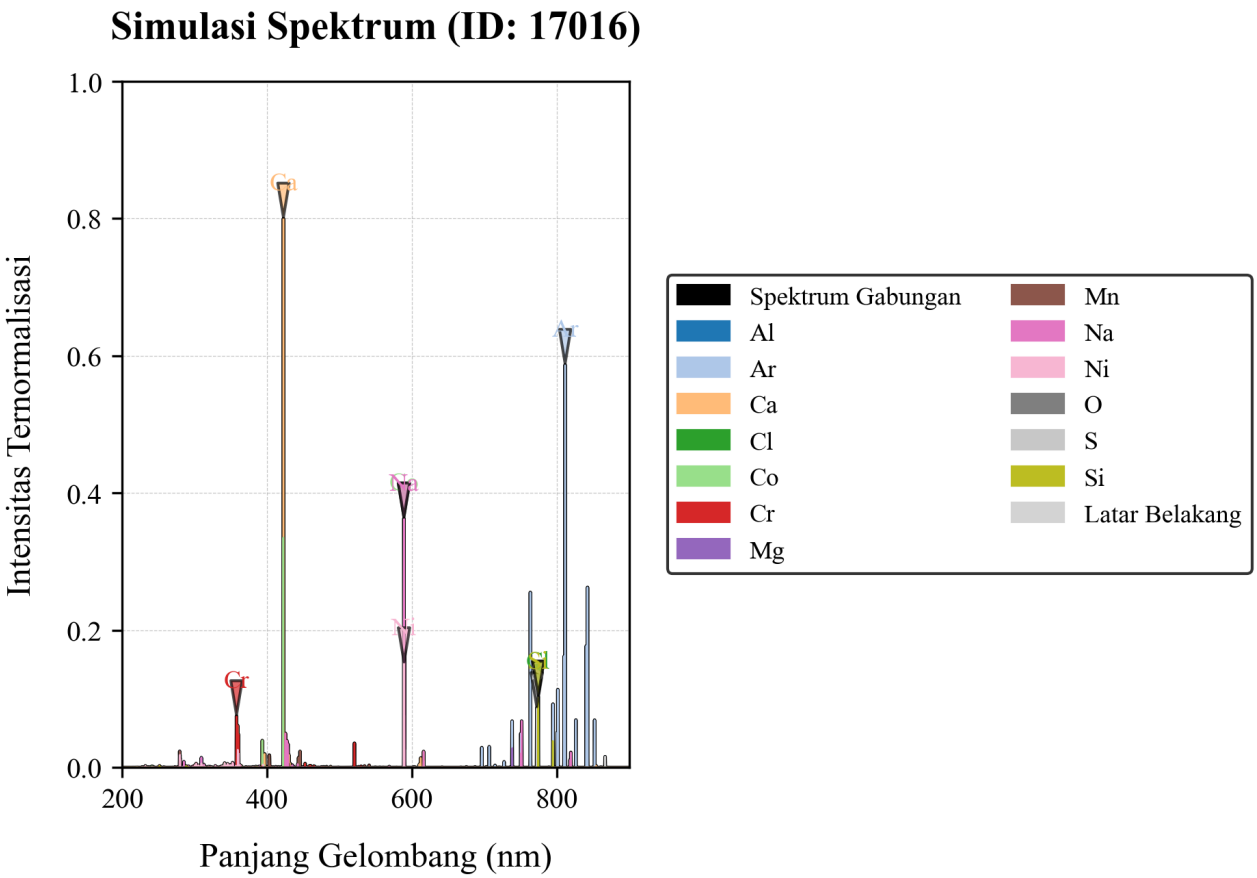

Figure 5: Spectral simulation plot for sample 17016 in full horizontal layout.